

Independent Validation Corpus - Order Fulfillment and Warehouse Management

COMPLETE MR / DECISION / FR AUTHORIZING PROBE - REVISION 3

This corpus replaces the two earlier simplified order/inventory holdouts for validation purposes. It is written against the regular MR -> Decision -> FR hierarchy. It is evidence/test material, not a universal domain reference architecture.

1. Problem framing

Problem / class of problems: enable customers to place orders for physical products and progress them to fulfillment while preserving coherent commitments across finite stock, concurrent demand, payment authorization and physical dispatch.

Meaningful boundary: from checkout/order evaluation through order commitment, warehouse stock movement/reservation, payment authorization/settlement and handoff to dispatch.

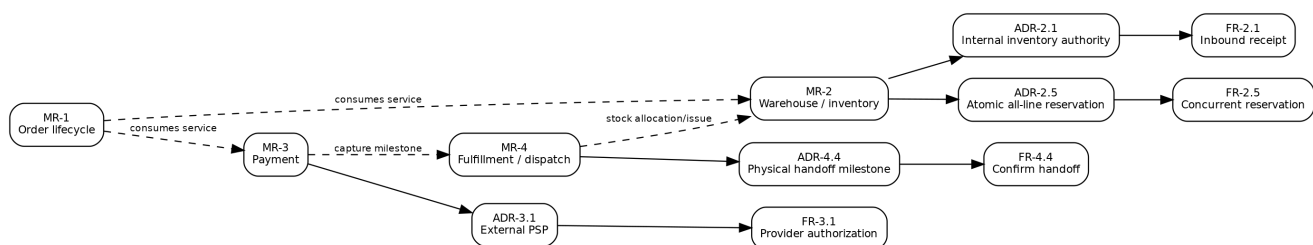
Out of scope: procurement planning, supplier replenishment, long-term demand forecasting and carrier transport after governed handoff.

Not assumed at framing level: local vs external WMS/ERP, payment provider, carrier, database, API protocol, message broker, microservices, cloud, reservation algorithm, warehouse topology or persistence technology.

2. Operating scenario

- products are physical and stock is finite;
- concurrent checkout sessions may request the same SKU;
- stock changes through receipts, issues, returns, damage/quarantine and controlled adjustments;
- orders may contain multiple lines;
- payment authorization may be authorized, declined or indeterminate;
- retries must not create duplicate financial operations;
- stock reservations are temporary until commit, release or expiry;
- fulfillment may encounter short picks/damage despite a prior reservation;
- carrier interaction can be accepted, rejected or indeterminate.

3. Macro project map



MR-1 Order lifecycle and commitment
| depends on services from
+--> MR-2 Warehouse and inventory management
+--> MR-3 Payment authorization and settlement
|
`--> MR-4 Fulfillment and dispatch (after acceptance)

MR-4 consumes stock-allocation/issue services from MR-2
MR-3 capture timing consumes a governed handoff milestone from MR-4
Service consumption does not transfer ownership.

4. Governed references used by FR clauses

These references are an authoring substrate only. Their final Base Analysis types are deliberately deferred.

Ref	Name	Probe kind	Meaning
ACT-Customer	Customer	actor	person submitting che
ACT-WarehouseOperator	WarehouseOperator	actor	receives/picks/packs p
ACT-InventoryManager	InventoryManager	actor	approves controlled ad
CMP-OrderService	OrderService	capability	owns order evaluation/
CMP-InventoryService	InventoryService	capability	project-side inventory
CMP-PaymentAdapter	PaymentAdapter	capability	project-side PSP integr
CMP-PaymentProvider	PaymentProvider	external capability	selected PSP
CMP-FulfillmentService	FulfillmentService	capability	pick/pack/handoff lifec
CMP-CarrierAdapter	CarrierAdapter	capability	project-side carrier inte
DAT-OrderEvaluation	OrderEvaluation	data	correlated checkout ev
DAT-Order	Order	data	governed committed/r
DAT-InventoryBalance	InventoryBalance	data	on-hand/reserved/bloc
DAT-StockMovement	StockMovement	data	receipt/issue/return/ad
DAT-AvailabilityResult	AvailabilityResult	data	sufficient/insufficient/in
DAT-Reservation	Reservation	data	temporary allocation/h
DAT-ReservationRequest	ReservationRequest	data	full-order stock hold re
DAT-ReservationResult	ReservationResult	data	reserved/unavailable/in
DAT-PaymentAuthRequest	PaymentAuthorizationRequest	data	amount/currency autho
DAT-PaymentAuthResult	PaymentAuthorizationResult	data	authorized/declined/in
DAT-PaymentSettlementResult	PaymentSettlementResult	data	capture/void/refund re
DAT-FulfillmentTask	FulfillmentTask	data	pick/pack work for acc
DAT-DispatchRequest	DispatchRequest	data	governed carrier hand
DAT-DispatchAcceptance	DispatchAcceptance	data	carrier accepted/reject
STORE-InventoryLedger	InventoryLedger	store candidate	chosen project-owned
STORE-ReservationStore	ReservationStore	store candidate	chosen persistence for
CON-PaymentContract	PaymentProviderContract	contract candidate	governed PSP interface
CON-CarrierContract	CarrierContract	contract candidate	governed carrier interf

5. MR-1 - Order lifecycle and commitment

Intent

Manage a coherent order lifecycle and determine when the project is allowed to make a customer commitment, using inventory and payment capabilities without absorbing their internal responsibilities.

Context

Checkout can fail because stock cannot be reserved, payment cannot be authorized or a dependent capability returns an indeterminate result. Commitment must not leave hidden stock/payment side effects.

Stakeholders

Customers, customer support, commercial/order operations, downstream inventory/payment/fulfillment owners.

Scope

Order evaluation/session, commitment decision, accepted/rejected/cancelled state and coordination/compensation of dependent capabilities. Warehouse accounting, PSP internals and carrier execution remain in their own branches.

ADR-1.1 - Explicit correlated OrderEvaluation before commitment

Context. The project could mutate a draft Order directly, keep checkout state externally/statelessly, or create an explicit evaluation context used to correlate stock/payment outcomes before commitment.

Decision. Use one explicit OrderEvaluation identity for each checkout evaluation. All reservation, payment and compensation results used for that attempt are correlated to it before an Order becomes committed.

Consequences. Correlation and idempotency become explicit project semantics; stale/superseded evaluations can be distinguished from the current attempt.

FR-1.1 - Create and preserve one correlated order evaluation

When Customer submits checkout with one or more product lines and quantities, OrderService MUST create exactly one active OrderEvaluation with stable correlation identity and MUST preserve the evaluated product/quantity set until the evaluation is terminal or superseded.

SPO:

Subject	Predicate	Object
Customer	submits	OrderEvaluation
OrderService	creates	OrderEvaluation
OrderService	maintains	OrderEvaluation

ADR-1.2 - Reservation and payment authorization are both commitment gates

Context. The project could confirm before payment, confirm before stock allocation, or require both results. Execution may be sequential or concurrent, but customer commitment must not exist without both prerequisites.

Decision. An Order may become accepted only after one valid full-order ReservationResult=reserved and one PaymentAuthorizationResult=authorized are correlated to the same active OrderEvaluation. The acquisition order/concurrency of the two results is not fixed here.

Consequences. MR-1 consumes capabilities from MR-2 and MR-3. Indeterminate results keep the evaluation non-committed. Side effects require compensation if the other gate fails.

FR-1.2 - Obtain governed inventory and payment results for the active evaluation

For an active OrderEvaluation, OrderService MUST request the required stock-reservation and payment-authorization services and MUST associate each returned governed result with the same evaluation before commitment is decided. A result correlated to another/superseded evaluation MUST NOT satisfy the current gate.

SPO:

Subject	Predicate	Object
OrderService	requests	InventoryService
OrderService	requests	PaymentAdapter
OrderService	receives	ReservationResult
OrderService	receives	PaymentAuthorizationResult

Service consumption: reservation is provided by the MR-2 branch; payment authorization by MR-3.

ADR-1.3 - All-or-nothing customer commitment for requested lines

Context. A multi-line order could accept available lines and backorder/reject others, or require the whole requested set before commitment.

Decision. The probe uses all-or-nothing customer commitment: every requested line/quantity must be covered by the reservation used for acceptance.

Consequences. Partial warehouse allocations cannot silently become a partially accepted customer order under this project policy.

FR-1.3 - Commit or reject the full evaluated order from correlated results

When both governed gate results are available for the same active OrderEvaluation, OrderService MUST transition/create Order=ACCEPTED only if payment is authorized and reservation covers the full evaluated line set. A negative result MUST prevent acceptance. An indeterminate result MUST keep the order non-committed until resolved or the evaluation is terminated by applicable recovery policy.

SPO:

Subject	Predicate	Object
OrderService	reads	ReservationResult
OrderService	reads	PaymentAuthorizationResult
OrderService	transitions	Order

ADR-1.4 - Explicit compensation of pre-commit side effects

Context. Reservation or payment authorization can succeed while the other gate later fails/cancels. Ignoring the successful side effect would leak stock holds or financial authorizations.

Decision. A non-accepted evaluation with created pre-commit side effects must explicitly invoke the corresponding release/void service and track unresolved compensation.

Consequences. Order failure is not treated as a single local calculation; compensation is a governed cross-MR interaction.

FR-1.4 - Compensate failed/cancelled commitment

When an OrderEvaluation terminates without acceptance after a stock Reservation or Payment Authorization side effect exists, OrderService MUST request the corresponding release/void capability and MUST retain enough correlation to determine the compensation outcome. It MUST NOT treat the evaluation as cleanly compensated while a required compensation result remains indeterminate.

SPO:

Subject	Predicate	Object
OrderService	requestsRelease	InventoryService
OrderService	requestsVoid	PaymentAdapter

6. MR-2 - Warehouse and inventory management

Intent

Maintain trustworthy operational stock and provide availability/reservation/issue behavior that prevents the project from promising the same finite quantity to incompatible commitments.

Context

Stock enters/leaves the warehouse, may be damaged/blocked, may be temporarily reserved by concurrent sessions and must eventually be consumed or released. Availability is not equivalent to the last counted on-hand quantity.

Stakeholders

Warehouse operators, inventory managers, order operations, fulfillment operations, finance/audit roles affected by stock discrepancies.

Scope

Accountable stock movements, stock-state semantics, availability, reservation lifecycle, concurrency and issue. Procurement/supplier planning is outside this probe.

ADR-2.1 - Project-owned internal inventory authority

Context. The project could manage stock internally, delegate stock/reservation to an external WMS/ERP, or consume a black-box reservation service. These alternatives move the responsibility boundary and materially change downstream FRs.

Decision. This concrete probe implements inventory management internally: InventoryService + InventoryLedger are the authoritative project-managed source for on-hand/reserved/blocked stock state used by ordering and fulfillment.

Consequences. The project owns receipt/adjustment/availability/reservation/concurrency/release/issue FRs. A future external-WMS Decision would replace many of these with integration/API/contract FRs rather than being a transparent implementation swap.

FR-2.1 - Record inbound stock receipt exactly once

When WarehouseOperator confirms receipt of a product quantity at a governed stock location, InventoryService MUST create an attributable inbound StockMovement and MUST increase corresponding on-hand InventoryBalance exactly once. Reprocessing the same governed receipt identity MUST NOT increase stock a second time.

SPO:

Subject	Predicate	Object
WarehouseOperator	confirmsReceipt	StockMovement
InventoryService	records	StockMovement
InventoryService	increases	InventoryBalance

Subject	Predicate	Object
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ADR-2.2 - Accountable stock changes are recorded as movements, not destructive balance rewrites

Context. Inventory corrections could overwrite current balance with little history, or preserve attributable movements from which current accountable state is updated/reconciled.

Decision. Receipt, issue, return and controlled adjustment are recorded as attributable StockMovement events; historical accountable movements are not silently rewritten/deleted to hide previous state.

Consequences. Reconciliation/auditability improve; corrections are additional governed movements. Future GovernanceRequirements may add approval/evidence constraints.

FR-2.2 - Record controlled stock adjustment

When a physical count, damage/quarantine event or approved correction requires an accountable change, InventoryService MUST create an adjustment StockMovement linked to the applicable reason/approval context and MUST update the affected InventoryBalance without deleting or rewriting the prior accountable movement history.

SPO:

Subject	Predicate	Object
InventoryManager	approves	StockMovement
InventoryService	records	StockMovement
InventoryService	updates	InventoryBalance

ADR-2.3 - Separate on-hand, reserved and blocked stock semantics

Context. One quantity cannot distinguish physically present stock from stock already promised to orders or present but unusable due to damage/quarantine.

Decision. Govern distinct on-hand, active-reserved and blocked quantities. Allocatable availability excludes active reservations and blocked quantity.

Consequences. Every movement/reservation/release/issue must preserve these semantics; exact persistence schema remains implementation.

FR-2.3 - Produce attributable current availability without freezing stock

When an availability query identifies product, location scope and requested quantity, InventoryService MUST derive AvailabilityResult from current accountable InventoryBalance, distinguish sufficient/insufficient/indeterminate and make the result attributable to its observation/state context. The availability query MUST NOT itself freeze quantity.

SPO:

Subject	Predicate	Object
InventoryService	reads	InventoryBalance
InventoryService	produces	AvailabilityResult

ADR-2.4 - Temporary expiring reservation hold bound to one OrderEvaluation

Context. The project could decrement stock immediately at checkout, use non-expiring allocations, perform best-effort availability checks without holds, or use temporary reservation/lease semantics.

Decision. Use a temporary Reservation bound to one OrderEvaluation. ACTIVE reservation freezes allocatable quantity without representing physical stock issue. It remains until commit/consumption, explicit release or governed expiry.

Consequences. Reservation identity/lifecycle/correlation become operational semantics. The expiry duration itself is not hard-coded into this FR corpus.

FR-2.4 - Maintain reservation identity and lifecycle correlation

For every ACTIVE Reservation, InventoryService MUST preserve exactly one originating OrderEvaluation reference, reserved line quantities, lifecycle state and applicable expiry context. Release, expiry or consumption commands MUST target that Reservation identity and MUST be idempotent for repeated delivery of the same governed command.

SPO:

Subject	Predicate	Object
InventoryService Reservation	maintains references	Reservation OrderEvaluation

ADR-2.5 - Atomic all-lines reservation under concurrency

Context. Concurrent checkouts can observe the same apparent availability and oversell if read/check and reservation are independent. Multi-line orders also risk partial holds that violate MR-1 all-or-nothing commitment.

Decision. Allocation for one full-order ReservationRequest is atomic with respect to competing reservations and stock-changing operations: either the complete requested set becomes reserved or no line becomes ACTIVE for that request.

Consequences. Implementation must provide concurrency control but the Decision does not prescribe locks, isolation level, CAS, queue or storage engine.

FR-2.5 - Create one atomic temporary reservation for the full evaluated set

When a ReservationRequest identifies an active OrderEvaluation and its full product/quantity set, InventoryService MUST atomically determine allocatability against current on-hand/reserved/blocked state. If fully allocatable, it MUST create exactly one ACTIVE Reservation, increase reserved quantity for each line exactly once and return ReservationResult=reserved. If not fully allocatable, it MUST create no active partial reservation for that request and return unavailable. A concurrent request MUST NOT reserve quantity already frozen by the first active Reservation.

SPO:

Subject	Predicate	Object
InventoryService	creates	Reservation
InventoryService	updates	InventoryBalance
InventoryService	produces	ReservationResult

ADR-2.6 - Reservation release/expiry restores allocatable quantity idempotently

Context. A temporary hold can be released explicitly, expire, or be consumed. Replayed lifecycle events must not double-release quantity.

Decision. Release and expiry transition an ACTIVE Reservation out of commitment eligibility exactly once and restore reserved quantity to allocatable state unless another stock constraint blocks it.

Consequences. Reservation lifecycle commands/results must be correlated and idempotent. The policy determining expiry time remains separately governed.

FR-2.6 - Release reserved quantity after cancellation, failure or expiry

When an ACTIVE Reservation is explicitly released because its order evaluation is cancelled/rejected, or when the governed expiry condition occurs, InventoryService MUST transition it so it can no longer support commitment and MUST decrease corresponding reserved quantities exactly once. Replayed release/expiry handling MUST NOT free the same quantity twice.

SPO:

Subject	Predicate	Object
InventoryService	releases	Reservation
InventoryService	decreasesReserved	InventoryBalance

ADR-2.7 - Physical handoff is the stock-issue milestone

Context. Accountable stock issue could be recorded at pick completion, packing, carrier request, carrier acceptance or physical handoff. Recording too early can reduce on-hand stock when goods remain controlled by the warehouse.

Decision. For this probe, accountable outbound issue occurs at the governed **physical handoff milestone** for the picked quantities, not merely at dispatch-request preparation or raw carrier acceptance.

Consequences. Fulfillment must emit/confirm the handoff milestone. Inventory consumes only quantities covered by the valid Reservation. Payment capture policy can reference the same governed milestone.

FR-2.7 - Convert reserved quantities into accountable stock issue at handoff

When FulfillmentService confirms the governed physical handoff milestone for an accepted order under a valid Reservation, InventoryService MUST create the attributable outbound StockMovement and MUST atomically reduce on-hand and reserved quantities for the consumed reservation lines exactly once. It MUST NOT issue quantities beyond the Reservation or silently consume unrelated unreserved stock.

SPO:

Subject	Predicate	Object
InventoryService	records	StockMovement
InventoryService	decreases	InventoryBalance
InventoryService	consumes	Reservation

6.1 Counterfactual: if ADR-2.1 chose an external WMS

This is not the chosen branch, but it demonstrates why ADR-2.1 is a real responsibility-boundary Decision.

With an external authoritative WMS, project-owned FRs would shift toward integration semantics such as:

- submit governed receipt/adjustment requests through the WMS contract;
- request full-order reservation and correlate the WMS result to OrderEvaluation;
- map provider-specific reservation statuses to governed project semantics;
- resolve indeterminate/time-out outcomes before semantically new retries;
- request release of the exact reservation identity;
- confirm/consume stock issue through the external contract.

The WMS's internal ledger schema, lock/isolation mechanism and internal concurrency algorithm would normally fall outside project FR ownership unless exposed as contractual properties.

Therefore "internal vs external inventory authority" is not an implementation detail: it changes the FR responsibility boundary.

7. MR-3 - Payment authorization and settlement

Intent

Provide governed payment authorization/settlement state that order commitment can consume without adopting provider-specific status as project semantics.

Context

External operations can time out after processing; blind retry can duplicate financial effects. Authorization, capture and void/refund are distinct lifecycle operations.

Stakeholders

Customers, finance, customer support, fraud/payment operations, order operations.

Scope

Authorization request/result, ambiguous outcome resolution, void/capture and provider integration. Card-network/provider internals remain outside project boundary.

ADR-3.1 - External PSP behind a governed project contract

Context. Payment could be processed internally or delegated to different providers with different APIs and state vocabularies.

Decision. Delegate payment operations to external PaymentProvider through PaymentProviderContract, with PaymentAdapter owning project-side integration.

Consequences. API interaction, correlation and failure handling are project FRs; provider internals are not.

FR-3.1 - Submit idempotently identifiable authorization request

When PaymentAuthorizationRequest is issued for an active OrderEvaluation, PaymentAdapter MUST submit an idempotently identifiable provider operation through PaymentProviderContract and preserve correlation to the evaluation.

SPO:

Subject	Predicate	Object
PaymentAdapter	sends	PaymentAuthorizationRequest
PaymentAdapter	communicatesWith	PaymentProvider

ADR-3.2 - Normalize provider-specific authorization states into project semantics

Context. Provider-specific status vocabularies should not leak into order commitment semantics.

Decision. Project payment authorization result is governed as authorized, declined or indeterminate; raw provider states are mapped behind PaymentAdapter.

Consequences. Provider replacement does not require MR-1 to understand provider-specific status codes.

FR-3.2 - Produce correlated governed authorization result

PaymentAdapter MUST map the provider response/outcome to PaymentAuthorizationResult in project semantics and MUST correlate it to the originating OrderEvaluation. A provider-specific raw status MUST NOT directly become the order commitment result.

SPO:

Subject	Predicate	Object
PaymentAdapter	maps	PaymentAuthorizationResult
PaymentAdapter	produces	PaymentAuthorizationResult

ADR-3.3 - Resolve indeterminate operation before semantically new retry

Context. A timeout can occur after the provider accepted/processed a request. A blind new authorization can duplicate financial effects.

Decision. For an indeterminate submitted operation, resolve the same idempotent operation status before starting a semantically new authorization unless the contract proves it was not accepted.

Consequences. Recovery queries/idempotency keys become operational contract; retry policy is not a naive loop.

FR-3.3 - Resolve indeterminate authorization safely

When authorization outcome is indeterminate after submission, PaymentAdapter MUST query/resolve the same provider operation before initiating a semantically new authorization attempt for that OrderEvaluation, unless the governed contract proves the original operation was not accepted.

SPO:

Subject	Predicate	Object
PaymentAdapter	queries	PaymentProvider
PaymentAdapter	updates	PaymentAuthorizationResult

ADR-3.4 - Capture only after governed physical handoff

Context. Payment could be captured at checkout, after picking, after carrier acceptance or after physical handoff.

Decision. Authorize before order acceptance, but capture only after MR-4 confirms the governed physical handoff milestone for the order.

Consequences. Accepted-but-not-handed-off orders hold authorization without capture; failed pre-handoff commitment requires void/release.

FR-3.4 - Capture authorized payment after handoff

When the governed physical handoff milestone is confirmed for an accepted order with still-valid authorization, PaymentAdapter MUST request capture exactly once for the governed payable amount and MUST produce a correlated PaymentSettlementResult. It MUST NOT capture solely because Order is ACCEPTED.

SPO:

Subject	Predicate	Object
PaymentAdapter	requestsCapture	PaymentSettlementResult
PaymentAdapter	produces	PaymentSettlementResult

ADR-3.5 - Void open authorization when pre-capture commitment is abandoned

Context. A reservation/payment gate can fail after authorization succeeds, or an accepted order can be cancelled before capture.

Decision. Open authorization is explicitly voided/released when commitment is abandoned before capture.

Consequences. Compensation is observable and correlated rather than left to provider expiration as hidden behavior.

FR-3.5 - Void pre-capture authorization

When an OrderEvaluation terminates without acceptance, or an accepted order is cancelled before capture while valid authorization remains open, PaymentAdapter MUST request the governed void/release operation and MUST produce a correlated PaymentSettlementResult.

SPO:

Subject	Predicate	Object
PaymentAdapter	requestsVoid	PaymentSettlementResult
PaymentAdapter	produces	PaymentSettlementResult

8. MR-4 - Fulfillment and dispatch

Intent

Transform an accepted order backed by valid stock allocation into controlled physical fulfillment and dispatch while preserving correlation between order, reservation, picked quantity, stock issue and carrier handoff.

Context

Picking can encounter shortage/damage despite reservation. Carrier request can fail/timeout. Physical stock must not be issued merely because a dispatch request exists.

Stakeholders

Warehouse/fulfillment operators, customers, carrier operations, customer support.

Scope

Fulfillment task, pick/pack confirmation, fulfillment exceptions, carrier request/acceptance and physical handoff confirmation. Carrier transport after handoff is outside scope.

ADR-4.1 - Fulfillment task requires accepted order plus valid reservation

Context. Warehouse execution could pick directly from order lines or require the exact stock allocation created during commitment.

Decision. Each FulfillmentTask references one accepted Order and the valid Reservation covering quantities to pick.

Consequences. Warehouse discrepancies can be detected against the committed allocation; inventory issue traces to reservation lifecycle.

FR-4.1 - Create fulfillment task from accepted order and reservation

When Order is ACCEPTED and its Reservation is valid for the committed line set, FulfillmentService MUST create exactly one active FulfillmentTask referencing both and containing quantities to pick. It MUST NOT create a normal pick task for rejected/non-committed evaluation.

SPO:

Subject	Predicate	Object
FulfillmentService	creates	FulfillmentTask
FulfillmentTask	references	Order
FulfillmentTask	references	Reservation

ADR-4.2 - Short pick/damage is an explicit exception, not silent substitution

Context. Physical execution may find missing/damaged quantity despite prior logical reservation.

Decision. Short pick/damage/missing allocated quantity produces an explicit fulfillment exception; normal completion cannot silently substitute unrelated stock commitment or mark full completion.

Consequences. Recovery/backorder/customer handling can be governed separately; stock/accounting is not silently corrupted.

FR-4.2 - Confirm picked quantities and surface reservation discrepancy

When WarehouseOperator confirms picking for a FulfillmentTask, FulfillmentService MUST record picked quantities by order line and compare them with the referenced Reservation. A short pick/damaged/missing allocated quantity MUST produce an explicit blocking fulfillment exception and MUST NOT be silently treated as complete fulfillment.

SPO:

Subject	Predicate	Object
WarehouseOperator	confirmsPick	FulfillmentTask
FulfillmentService	updates	FulfillmentTask

ADR-4.3 - External carrier handoff behind governed contract

Context. Dispatch could be performed internally or delegated to an external carrier/fulfillment provider.

Decision. Delegate carrier request through CarrierAdapter + CarrierContract; expose governed DispatchAcceptance rather than raw provider status.

Consequences. Provider mapping/correlation/recovery are project concerns; carrier transport internals remain external.

FR-4.3 - Submit complete dispatch request and produce governed acceptance

When FulfillmentTask has required picked/packed quantities and no blocking exception, CarrierAdapter MUST create/submit the governed DispatchRequest through CarrierContract and MUST produce DispatchAcceptance correlated to Order and FulfillmentTask. Raw carrier status MUST NOT directly become project dispatch state.

SPO:

Subject	Predicate	Object
CarrierAdapter	sends	DispatchRequest
CarrierAdapter	produces	DispatchAcceptance

ADR-4.4 - Explicit physical handoff milestone closes warehouse custody

Context. Carrier acceptance of a request does not necessarily mean goods physically left warehouse control. Stock issue/payment capture need a stable project milestone.

Decision. Define a governed physical handoff milestone after successful prepared dispatch, distinct from mere request acceptance. This milestone is the event consumed by MR-2 stock issue and MR-3 capture policies.

Consequences. MR-2 and MR-3 can depend on one governed business event without adopting raw carrier semantics.

FR-4.4 - Confirm physical handoff and invoke downstream services

When picked/packed goods for the accepted order physically cross the governed handoff milestone, FulfillmentService MUST record/emit that handoff correlated to Order, FulfillmentTask and Reservation, and MUST request the MR-2 stock-issue capability. The MR-3 capture branch may consume the same handoff result under ADR-3.4. A rejected/indeterminate carrier request alone MUST NOT count as physical handoff.

SPO:

Subject	Predicate	Object
FulfillmentService	confirmsHandoff	FulfillmentTask
FulfillmentService	requestsStockIssue	InventoryService

9. Hierarchy view

- MR-1 Order lifecycle
 - ADR-1.1 explicit evaluation
 - FR-1.1 create evaluation
 - ADR-1.2 commit gates
 - FR-1.2 obtain/correlate stock+payment results
 - ADR-1.3 all-or-nothing commitment
 - FR-1.3 commit/reject full order
 - ADR-1.4 compensation
 - FR-1.4 release/void side effects
- MR-2 Warehouse/inventory
 - ADR-2.1 internal inventory authority
 - FR-2.1 inbound receipt
 - ADR-2.2 accountable movement history
 - FR-2.2 controlled adjustment
 - ADR-2.3 stock-state semantics
 - FR-2.3 availability
 - ADR-2.4 temporary reservation lifecycle
 - FR-2.4 reservation identity/lifecycle
 - ADR-2.5 atomic all-line allocation
 - FR-2.5 concurrent reservation
 - ADR-2.6 release/expiry semantics
 - FR-2.6 release quantity
 - ADR-2.7 physical handoff -> stock issue
 - FR-2.7 outbound issue
- MR-3 Payment
 - ADR-3.1 external PSP
 - FR-3.1 submit authorization
 - ADR-3.2 normalized payment semantics
 - FR-3.2 governed result
 - ADR-3.3 indeterminate retry semantics
 - FR-3.3 resolve before retry
 - ADR-3.4 capture after handoff
 - FR-3.4 capture
 - ADR-3.5 void abandoned authorization
 - FR-3.5 void
- MR-4 Fulfillment
 - ADR-4.1 accepted order + reservation
 - FR-4.1 fulfillment task
 - ADR-4.2 explicit pick exception
 - FR-4.2 pick/discrepancy
 - ADR-4.3 external carrier contract
 - FR-4.3 dispatch acceptance
 - ADR-4.4 physical handoff milestone
 - FR-4.4 confirm handoff

No necessity/default Decision is required in this particular corpus because material alternatives exist for every parent commitment. The rule remains available for rare singleton-solution branches.

10. Validation observations

This corpus supports the regular hierarchy without requiring Decisions below FR:

- responsibility-boundary choices (internal WMS vs external WMS, external PSP, external carrier) sit naturally at Decision level;
- strategy/policy choices (reservation semantics, atomic all-line allocation, handoff milestone, capture timing) sit naturally at Decision level;
- operational obligations become concrete FRs under those commitments;
- cross-MR composition appears as service/capability consumption rather than multiple FR owners;
- no FR needs to own a Decision child;
- mutable values such as reservation duration remain outside FR unless later classified by a specialized/policy/interface contract.

The exact formal object used to represent cross-MR service consumption remains deliberately open for the Base Analysis phase.